



Fig. 5: Networking is bringing together all parts of the enterprise jigsaw (Courtesy: Supertech Instrumentation)

more powerful yet simple, thanks to wireless.

Wireless technologies make the communication setup within campuses quite easy and cost-effective. Initial installation and maintenance of wireless networks is comparatively inexpensive because one does not need to lay wires or replace them in the long run. Plus, adding new devices to a wireless network is also easy. Quite obviously, one can reach even difficult areas through wireless. This is especially useful in the case of sensor networks.

Perhaps one of the most important benefits of wireless networks is the reliability that is achieved by combining meshing with spread-spectrum technology. While doubts about security continue to nag manufacturers, the truth is that if done right, encryption, frequency differentiation, and techniques like direct-sequence spread-spectrum (DSSS) can surely ensure tight security for industrial wireless networks.

Several technologies such as ZigBee, ultra wideband (UWB), Wi-Fi, Bluetooth and WirelessHART are contending to rule the industrial automation space, while standards such as the ISA100 are attempting to streamline things.

Efforts are constantly on to make wireless the *de-facto* choice for industrial automation and to improve interoperability between various technologies. In May 2009, for example, the DASH7 Alliance was formed to create wireless technology that extends the ISO 18000-7 standard for low-power wireless data transfer.

The group aims to improve interoperability between radio frequency identification (RFID) devices and applications, which will, in turn, improve tracking of materials and cargo in the supply chain. Soon, DASH7 will align itself with complementary technologies like cellular, passive RFID,

WiFi, and 2-D barcode.

4 Smart devices can talk to each other, and even make decisions

Almost all industrial automation equipment are becoming smart nowadays, acting as standalone units without any dependence on PCs. In a modern factory floor, a device can store a wealth of information such as when it was installed and by whom, uptime and downtime, critical specifications, diagnostics, availability of spares, replacement alternatives, repair instructions, usage patterns and more. With a wee bit of intelligence, it can also update this information to reflect otherwise invisible machine activity. Some advanced systems may include decision-making technologies such as operations research and artificial intelligence, which control work flow, material and information flow, etc.

Add to this the power of networking, and you have smart devices that can share information, instructions and decisions with other devices as well as organisation-wide information systems.

The ability of today's machines to communicate with each other is called machine-to-machine (M2M) communication, and is one of the main benefits of wireless technology on the plant floor. M2M communication makes the information stored and generated by a machine more useful. It makes the information about assets, costs, liabilities and activities more visible to managers and to the decision-making process.

The effect is an unprecedented level of productivity and efficiency.

"Increased intelligence of industrial automation and 'smart' inspection devices has led to systems which are more compact and cost-effective. This has also led to more flexible systems which can be reconfigured easily to quickly handle changing automation demands, reducing the total cost of ownership (TCO)," says Yashasvi Nathan, senior engineer-Marketing, Soliton Technologies.

"We can take an example of vision-based inspection systems to illustrate this. In conventional PC-based vision inspection systems, the images from a camera would be fed to a processor. The image processing would be carried out on this separate processing platform (a PC typically). This would lead to bulky inspection systems. With smart cameras, however, the processor and camera are integrated into a single, highly-compact package. This is a completely standalone inspection unit which does not require any external processor for running the inspection algorithm. This system can be set up with minimal disturbance to the existing equipment. The standalone nature of this system makes it a very useful product. These smart cameras come with network connectivity, enabling decision makers to get information online for quick and appropriate decision-making."

Smart devices, pervasive computing and M2M communications are all set to change the role and scope of industrial automation, representing a huge business opportunity for industrial automation equipment suppliers. Their job will no longer be to supply mere machines. In order to tap a larger market, they will have to start looking at each machine as part of a larger system, and bundle the equipment with so-called smart services.

5 Integration gives more choice to customers

Networking, standards, convergence of communication technologies, powerful